BACTERIOPHAGES

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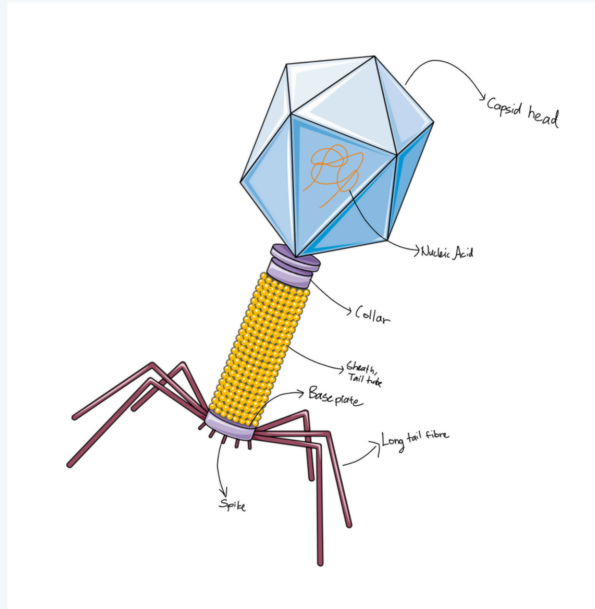
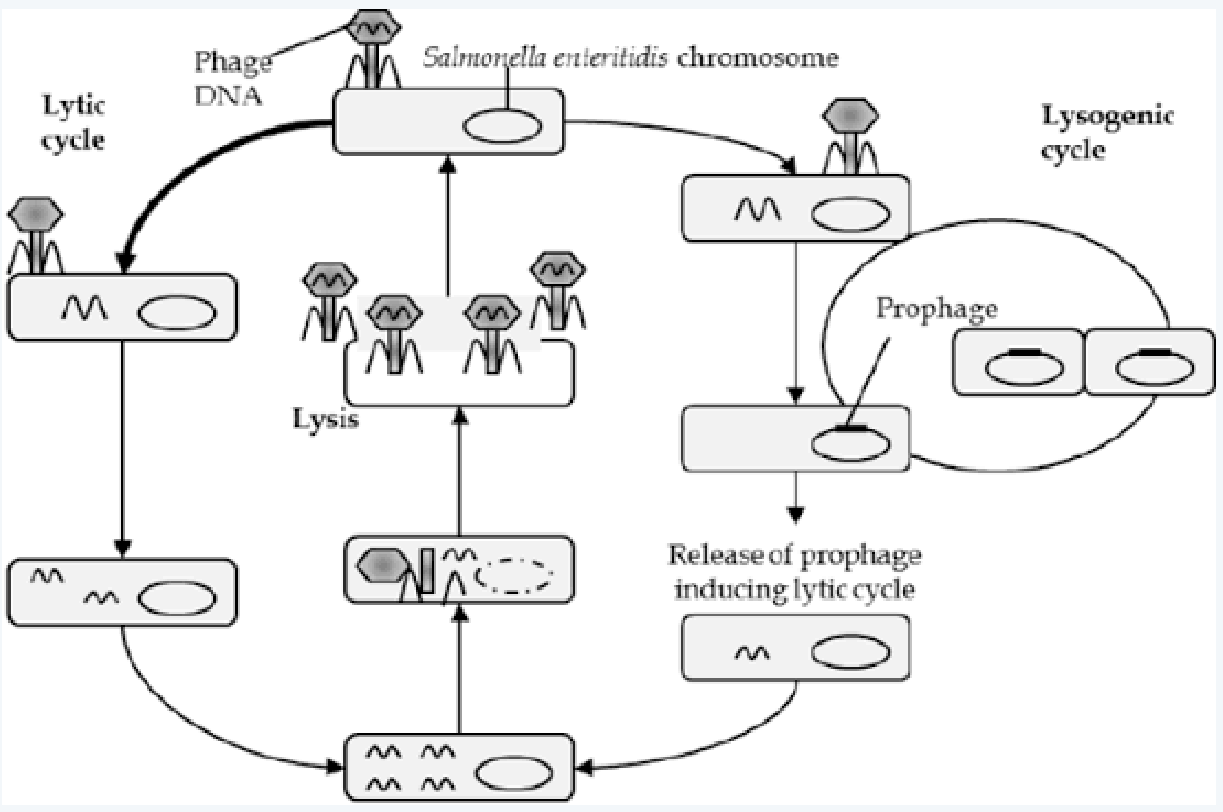


WHAT ARE THEY ?

- Bacteriophages are a group of viruses that infect bacteria
- They control the microbiome diversity of the environment by selecting for phage-resistant bacteria.

HOW DO THEY KILL BACTERIA ?

- Adhesion to the bacteria occurs , followed by the release of genetic material into the bacteria
- Disruption of metabolism occurs next and the bacteria's resources are used to form more viruses.
- The host cell, overwhelmed by the sheer number of viruses, bursts to release them.



BACTERIOPHAGES: THE SILENT KILLER

- Bacteriophages have a distinct alternative cycle other than the lytic cycle (mentioned above) known as the lysogenic cycle.
- Integration of bacteriophage genome into the bacteria's genome occurs upon entry.
- The host then replicates with the bacteriophage's genetic material.
- Bacteriophages may exit the lysogenic cycle and enter the lytic cycle at any time. (see above for lytic cycle)

APPLICATIONS

- Phage therapy opens up various avenues and applications, with regards to antimicrobrial resistance.
- In gene therapy, bacteriophages can be used as a vector to be injected inside our bodies.
- Engineering phages are modified to enhance their abilities to create more tools such as delivering drugs and combating against infection as well as preventing it.